

Reasoning Chain-of-Thought

Safety deliberation

CP

Harmful generation

Extract hidden states ↓ *Select optimal layer*

Crossing-rate composite

Threshold-based metrics

Shallow layer (L2, 3%)

Deep layer (L63, 98%)

✗ $p = 0.83$

(non-significant)

✓ $p < 0.005$

(significant)

Same data, same model (QwQ-32B)

Metric choice alone reverses the conclusion